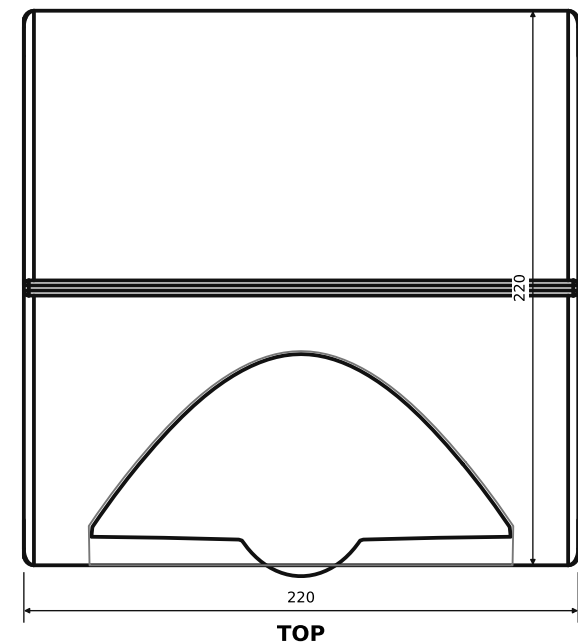
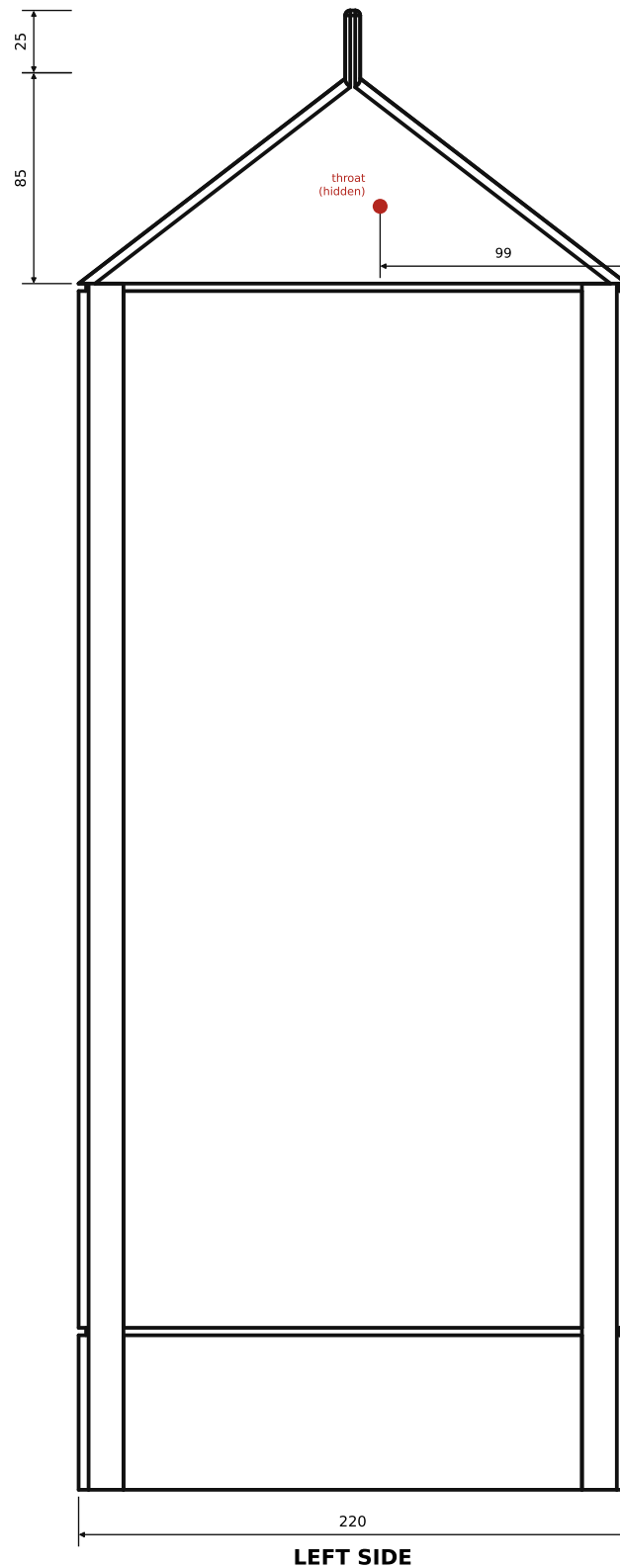
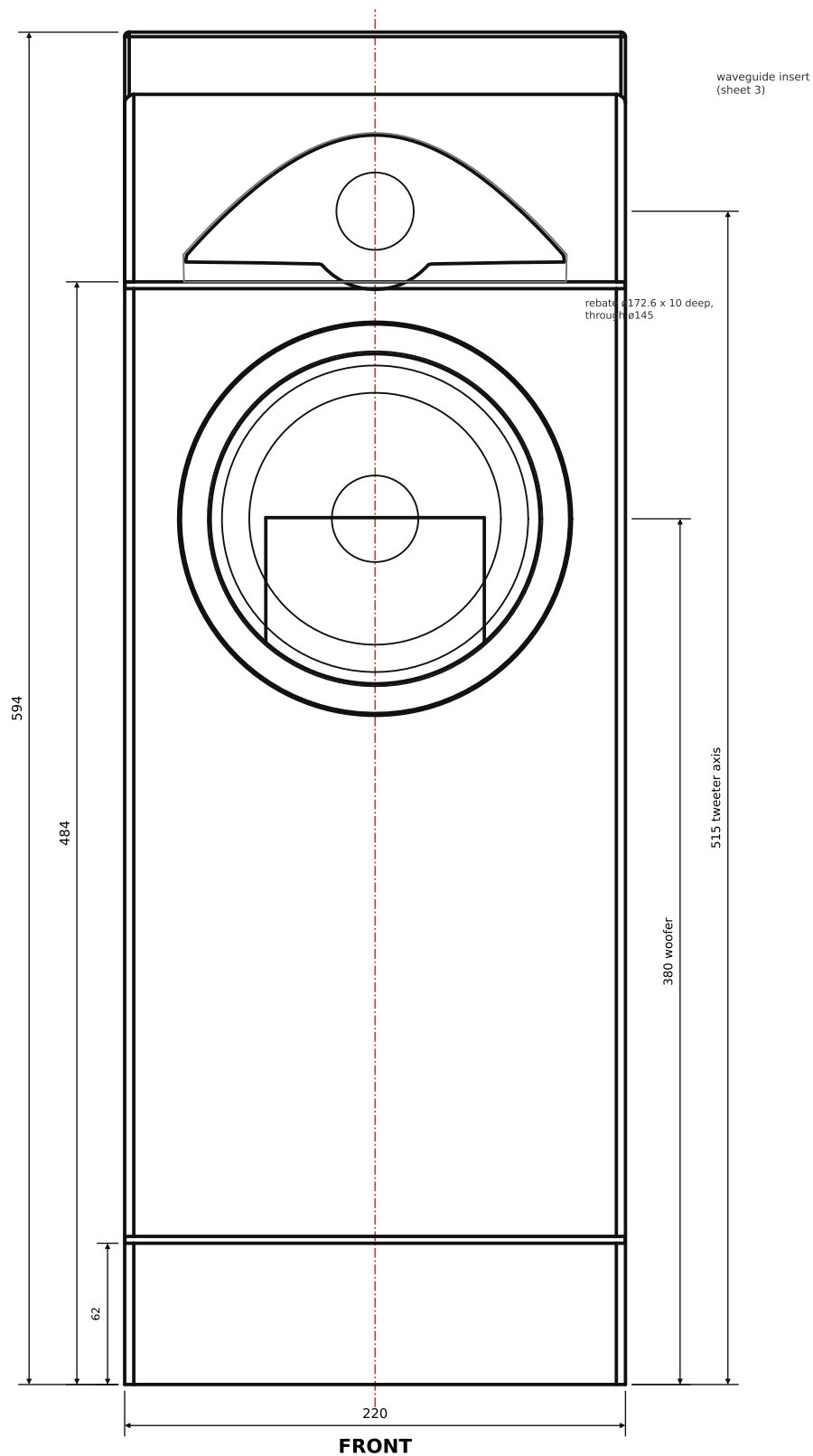




black: the waveguide's mouth, where its lip meets the roof
grey: the insert's seam

Notes

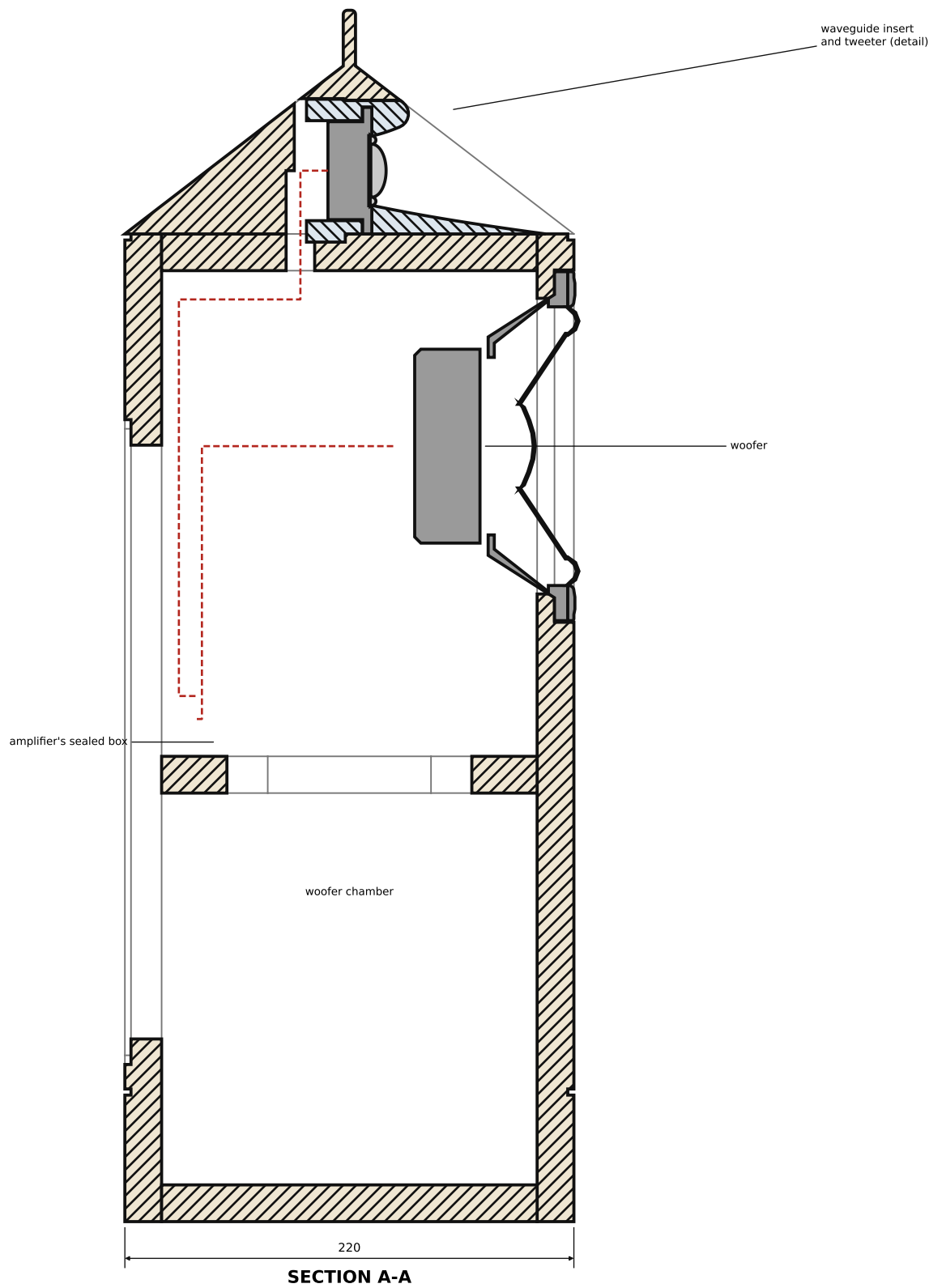
1. Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 6 mm, the fin's edges 3 mm. 3 x 3 mm shadow lines at z 110 and z 857.
2. The drivers sit flush: each frame in a rebate, a 3 mm trim ring over its flange and screws, level with the finish.
3. The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 515, 99 behind the front face. The shape is set by simulation (fab/out/acoustics/waveguide).
4. The back (port, amplifier, label) is drawn on sheet 4 with the wiring.
5. Finish: colour coat and 2K clear; the Facts printed between them.

earmilk bookshelf | General arrangement

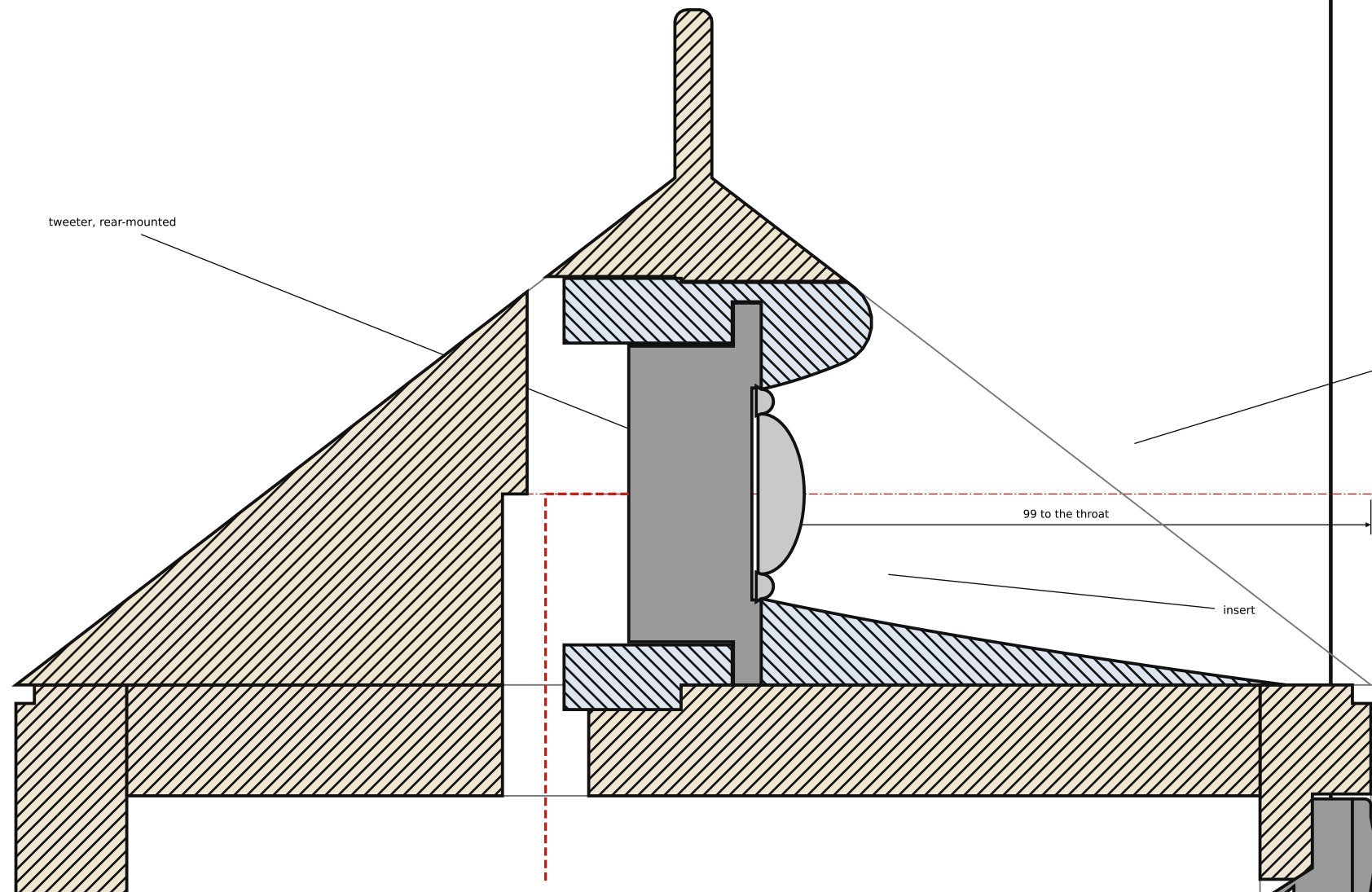
Front, left side and top at 1:3. Heights from the floor; depths from the front face.

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red dashed: the cables' runs to the amplifier (sheet 4)



The roof

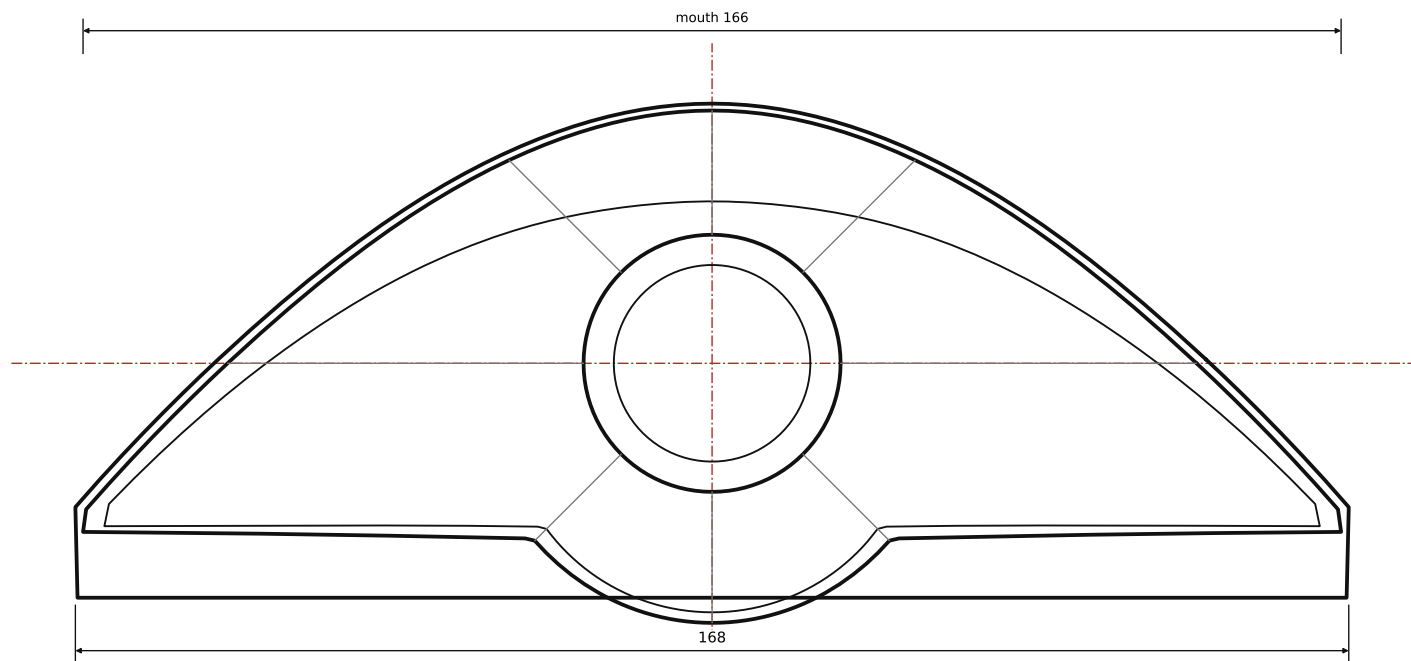
1. The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round. It slides out forward, level, with the tweeter on it.
2. The tweeter is rear-mounted: its flange in a counterbore behind the throat, its dome and surround filling the 34 mm throat, so the wall runs on from the surround with no step.
3. Its cable leaves through the boss's open back into a 14 mm channel down through the block and the top panel, to a connector in the woofer chamber with slack for the insert's travel.
4. Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).

earmilk bookshelf | Section on the centreline

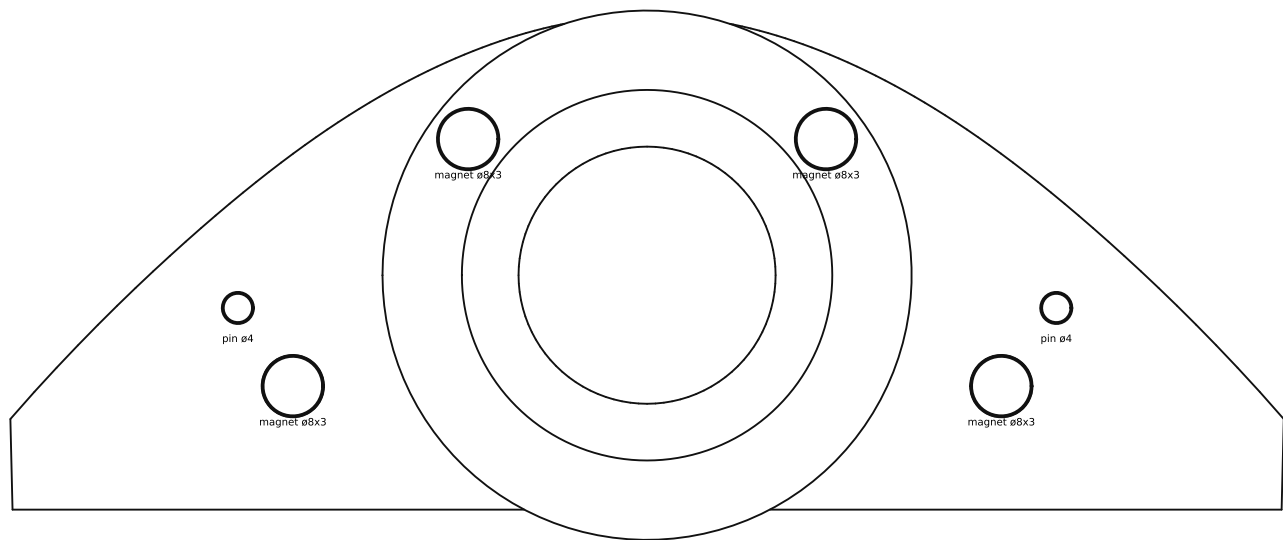
Section A-A on the centre plane (x 110) seen from the left at 1:3, and the roof at 1:1. Cut wood hatched; the insert hatched the other way.

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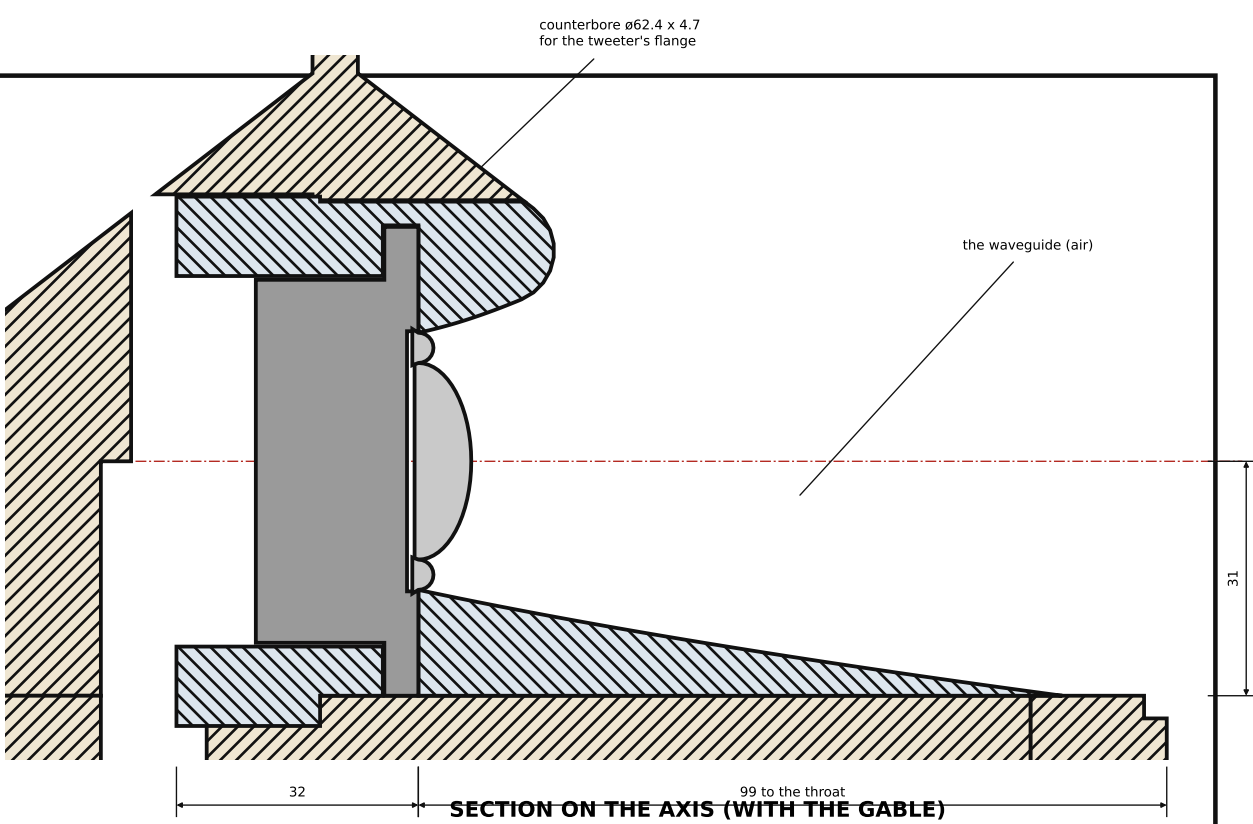
Check every bought part against its own drawing before cutting.



FRONT



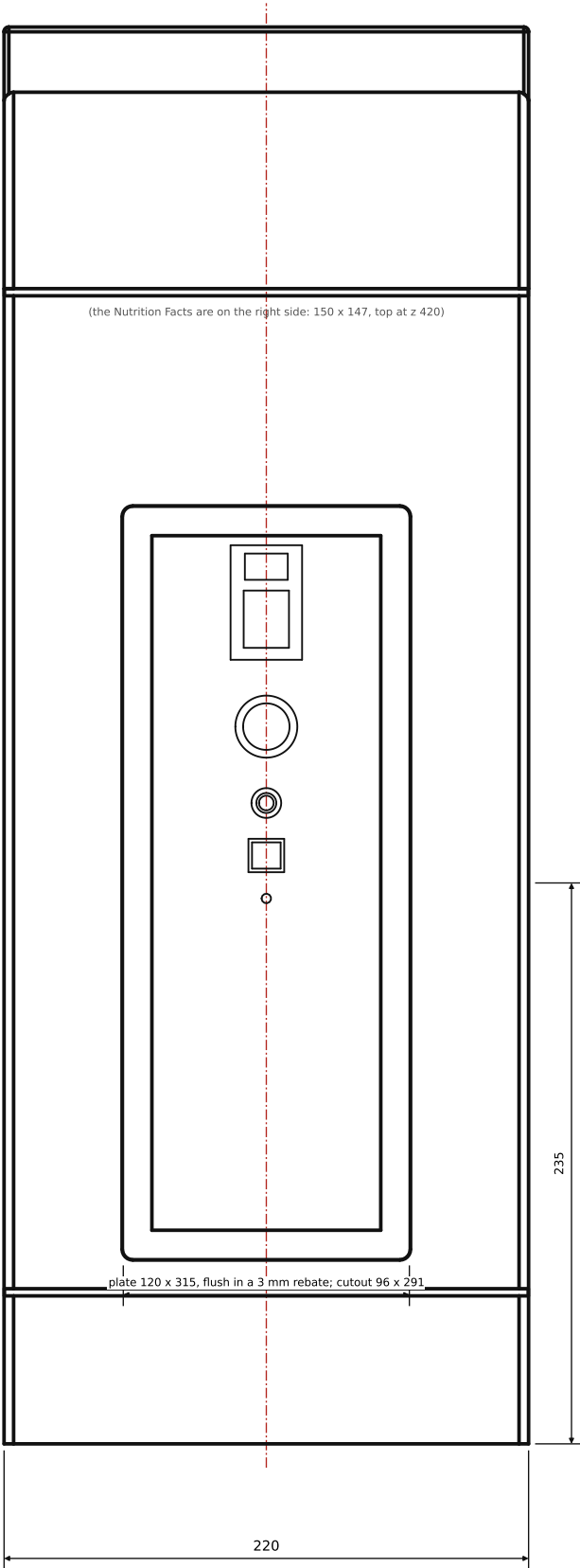
BACK (THE FACE THAT MEETS THE POCKET)



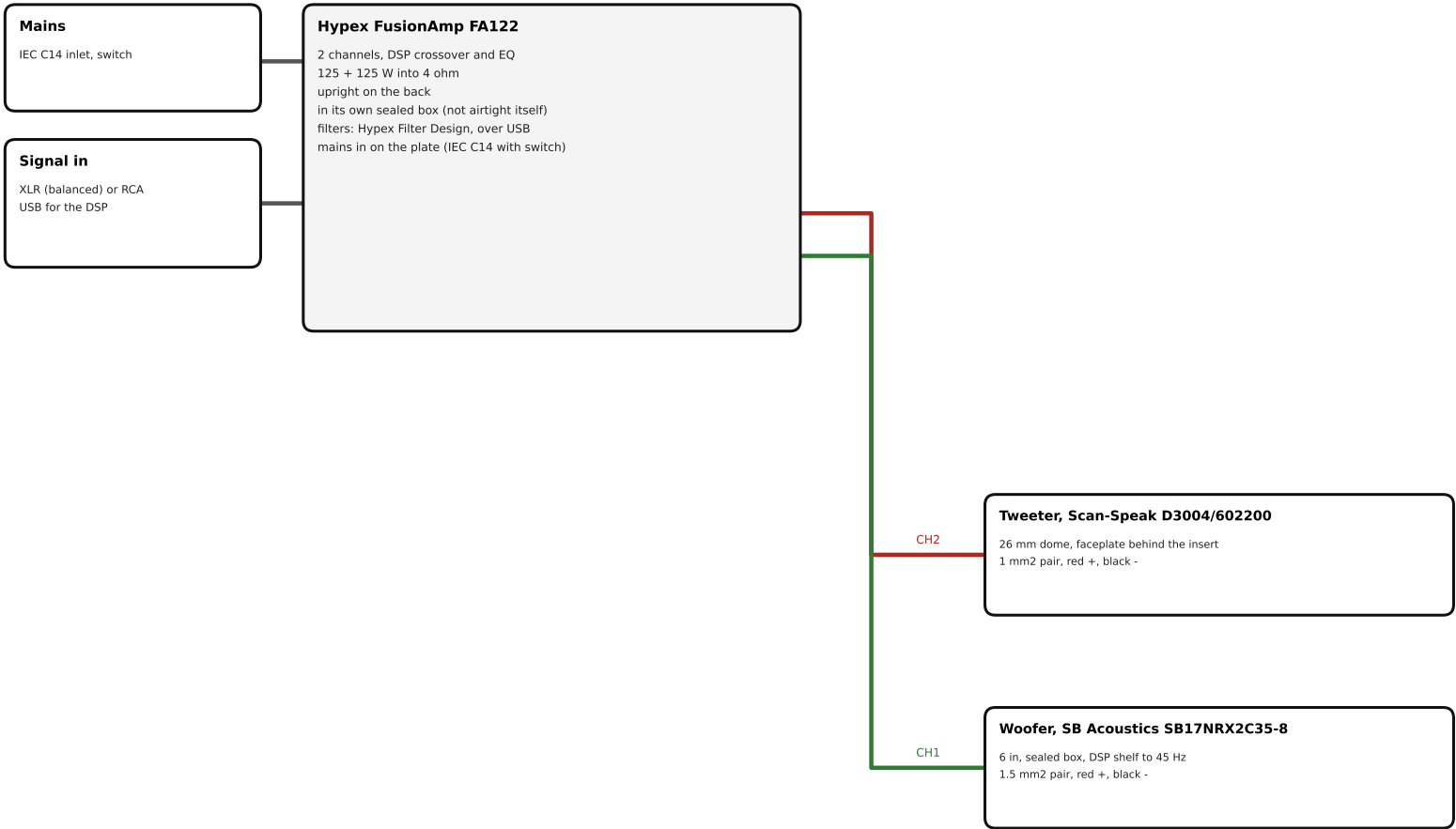
SECTION ON THE AXIS (WITH THE GABLE)

Fitting the tweeter and the insert

1. Solder the tweeter's lead (2 x 1.5 mm², 450 mm) to its tabs and fit the plug half of the connector. Feed the lead through the boss from the front.
2. Seat the tweeter in the counterbore from behind, dome toward the throat, on a 0.5 mm foam gasket. Fix its 3 screws into the insert (heat-set brass inserts, M3).
3. Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Fit the two pins in the insert.
4. From the front, feed the lead into the pocket's channel, plug it into the socket on the cabinet's lead (pulled up through the channel from the woofer chamber), push the slack back down.
5. Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even.
6. Service: lift it out with a suction lifter on its flat border (or a pull loop under its lower edge), unplug, and the tweeter comes out with it.



BACK



Runs, connectors and seals

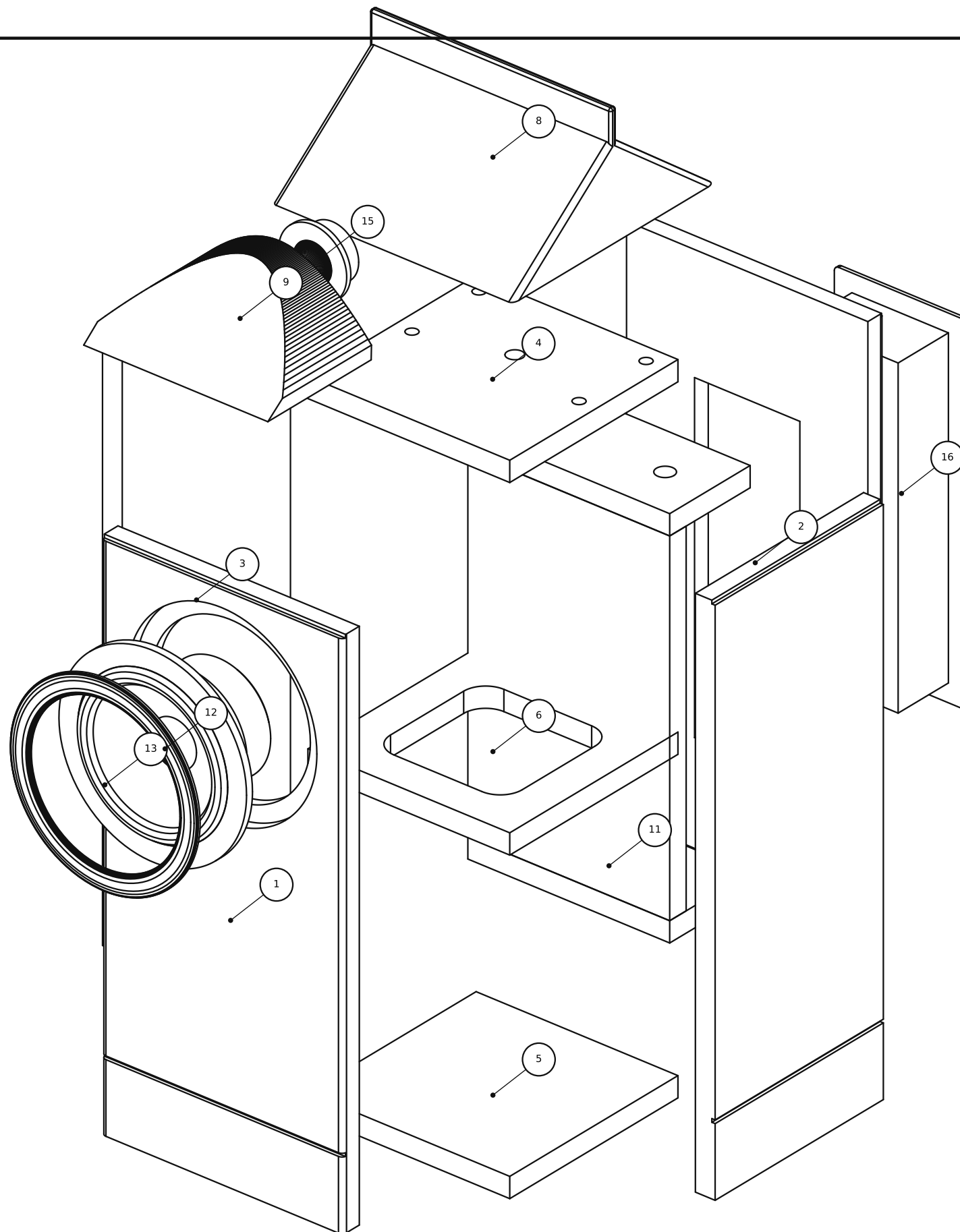
1. Woofer (CH1): from the amplifier box's gland to the woofer's terminals, 0.4 m of 1.5 mm2.
2. Tweeter (CH2): up the back wall to the top panel, through the 14 mm channel in the top panel and the gable block, 0.6 m of 1.0 mm2, to a two-pole locking connector at the insert.
3. Seal the tweeter cable where it passes the top panel (grommet and sealant); the box is sealed, and the insert's pocket is open to the room through its seam.
4. The amplifier box's lid: a 16 mm gland for the two pairs; seal it.
5. The crossover (about 2 to 2.5 kHz), the woofer's shelf to 45 Hz (+7.6 dB) and the delays are set in the DSP. Wire red to + throughout.

earmilk bookshelf | Back and wiring

The back at 1:3 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.

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Parts

- 1 **Front baffle x1**
18 mm Baltic birch, CNC: the woofer's cutout and rebate
- 2 **Back x1**
18 mm birch: the amplifier's cutout and rebate
- 3 **Sides x2**
18 mm birch; the Facts printed on the right one
- 4 **Top panel x1**
18 mm birch, the tweeter cable's 14 mm hole, four dowel holes
- 5 **Bottom panel x1**
18 mm birch
- 6 **Window brace x1**
18 mm birch
- 8 **Gable block x1**
laminated birch or printed whole: the insert's pocket and the cable channel
- 9 **Waveguide insert x1**
printed (SLA, MJF) or CNC from solid; magnets, pins
- 11 **Amplifier box x3**
18 mm birch floor, lid (gland) and front, glued and sealed
- 12 **Woofer x1**
SB Acoustics SB17NRX2C35-8, 6 in coated paper
- 13 **Trim ring x1**
printed, satin black or gunmetal, over the frame and its screws
- 15 **Tweeter x1**
Scan-Speak Illuminator D3004/602200, faceplate behind the insert
- 16 **Amplifier x1**
Hypex FusionAmp FA122, DSP, 2 channels

earmilk bookshelf | Exploded view and parts

The parts pulled apart along the way they go together, and what each one is.

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